

The Sharp Logarithmic Block Clock for Hilbert–Schmidt Spectral Tails

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July 2026

Abstract

We determine the exact logarithmic clock forced by a Hilbert–Schmidt mass law. Let a normal diagonal spectral tail have eigenvalue moduli at most q and squared mass $M_\sigma \leq C\sigma^{-\alpha}$. On a disk of radius R with $qR < 1$, a clock $m_\sigma = \lceil a \log(1/\sigma) \rceil$ suppresses the high-order regularized-determinant tail when $a \log(1/(qR)) \geq \alpha$. Equality gives only a logarithmic gain. Below that boundary a diagonal saturation family makes the tail diverge, so the threshold is sharp for this information class. The mass-and-cap root-rate upper bound is strictly below one under strict inequality, but a particular nonsaturating family may satisfy the RH-279 test at a smaller slope.

1 Mass-and-cap class

Let $C_\sigma = \text{diag}(\mu_j(\sigma))$ with $|\mu_j(\sigma)| \leq q$ and

$$M_\sigma = \sum_j |\mu_j(\sigma)|^2 \leq C\sigma^{-\alpha}, \quad \alpha > 0.$$

For $qR < 1$ and $m \geq 2$, the same estimate as in RH-282 gives

$$T_\sigma(m, R) := \sum_{n \geq m} \frac{|\text{Tr } C_\sigma^n| R^n}{n} \leq \frac{Cq^{-2}\sigma^{-\alpha}(qR)^m}{m(1 - qR)}. \quad (1)$$

2 Sharp clock theorem

Theorem 2.1. *Put $m_\sigma = \lceil a \log(1/\sigma) \rceil$. If*

$$a \log \frac{1}{qR} > \alpha,$$

then $T_\sigma(m_\sigma, R)$ decays by a positive power of σ . If equality holds, the right side of (1) is $O(1/\log(1/\sigma))$ and still tends to zero.

If $a \log(1/(qR)) < \alpha$, there exists a diagonal family satisfying the same mass and cap hypotheses for which $T_\sigma(m_\sigma, R) \rightarrow \infty$.

Proof. The upper assertions follow after writing $(qR)^{m_\sigma} = O(\sigma^{a \log(1/(qR))})$ in (1). For sharpness take $N_\sigma = \lfloor c\sigma^{-\alpha} \rfloor$ copies of q , with $c > 0$ small enough that $N_\sigma q^2 \leq C\sigma^{-\alpha}$. Then

$$T_\sigma(m, R) = N_\sigma \sum_{n \geq m} \frac{(qR)^n}{n}.$$

The first term gives $T_\sigma \geq N_\sigma(qR)^m/m$, which diverges when the displayed strict reverse inequality holds. $\square$

Proposition 2.2 (uniform root-rate boundary). *For the saturation scale $M_\sigma \asymp \sigma^{-\alpha}$,*

$$\limsup_{\sigma \downarrow 0} \|C_\sigma^{m_\sigma}\|_1^{1/m_\sigma} R \leq qRe^{\alpha/a}.$$

The right-hand side is strictly smaller than one exactly when $a > \alpha/\log(1/(qR))$. Thus the mass-and-cap information uniformly guarantees the RH-279 root-rate test in the strict supercritical region. For the repeated- q saturation family used above, the actual root-rate limit equals $qRe^{\alpha/a}$, so this boundary is exact for that family.

Proof. Use $\|C^m\|_1 \leq M_\sigma q^{m-2}$ and take m th roots. For $N_\sigma \asymp \sigma^{-\alpha}$ repeated entries equal to q , $\|C_\sigma^{m_\sigma}\|_1 = N_\sigma q^{m_\sigma}$, which gives equality in the limiting root rate. $\square$

3 Hardy-disk instance and boundary

For $\alpha = 1$, $q = 1/2$, and $R = 7/5$,

$$a_{\text{crit}} = \frac{1}{\log(10/7)} = 2.803673252\dots$$

The RH-282 choice $a = 4$ has strict exponent $4\log(10/7) - 1 = 0.426699775\dots$

The lower model proves sharpness only from mass and modulus information. It does not claim that the folded Gaussian spectrum has repeated roots at q . The spectral-to-monodromy bridge and Gates A–E remain open.